WALKABILITY IN EDUCATION CITY

Addressing Pedestrian Comfort Through Researched Solutions

Team Cocktail

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THE PROBLEM: WALKING IN THE HEAT

40°C+
Summer temperatures (Visit Qatar, 2026).

57%
Drive private vehicles (Al-Mohannadi et al., 2023).

17%
Walk between facilities (Al-Mohannadi et al., 2023).

Why It Matters

- EC spans 12 km² with 24 tram stations, yet extreme heat and unshaded walkways make walking unbearable.
- Students, faculty members, staff, and visitors are most affected.
- Increased carbon emissions, reduced physical activity.
- Contradicts with QNV 2030 targets

QNV 2030 targets: 25% emissions reduction • 60–70% pedestrian shade • Sustainable transport

SOLUTIONS EVALUATED

1. Park and Plant Initiative

Native Sidra & Ghaf trees along walkways to create shaded corridors

75% canopy

optimal airflow + cooling

Limitation: 3-5 years to full canopy maturity

Plant a million trees by 2030 initiative by Ministry of municipality

Development of public space like Al Bidda park for enhanced walkability

Visualization of ghaf trees providing a shaded walkway within Education City.

SOLUTIONS EVALUATED

2. Enclosed Walkways

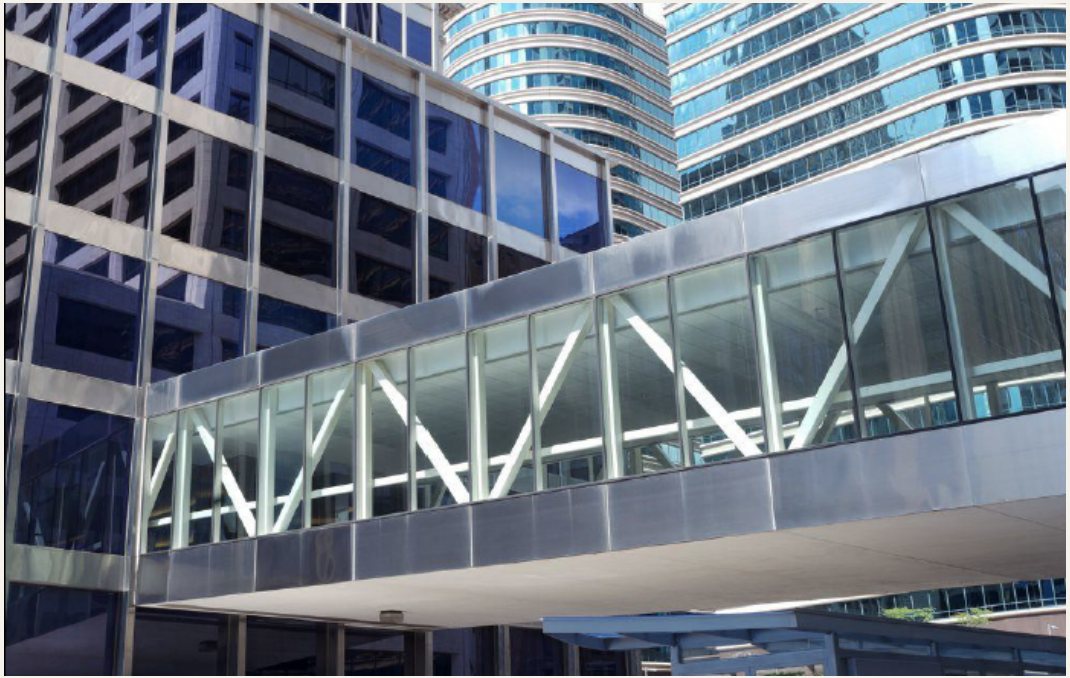
Air-conditioned corridors between buildings

Full A/C

complete climate control

Limitation

High energy + cost; poor fit for EC's layout



External view of an enclosed glass-and-steel skyway bridge connecting urban commercial buildings

SOLUTIONS EVALUATED

3. Solar-Powered Misting

Solar-powered misting nozzles under canopies

−12.3°C

thermal index (ideal humidity)

Limitation

Drops sharply above 60–70% humidity

Visualization of a misting system on a patio

WEIGHTED DECISION MATRIX

Criteria	Weight	Park & Plant	Cool Pavement	Enclosed Walkways	EC Canopies (Reference)
Thermal Comfort	30%	**3 (0.90)**	1 (0.30)	5 (1.50)	3 (0.90)
Sustainability	30%	**5 (1.50)**	2 (0.60)	1 (0.30)	3 (0.90)
Coverage & Scalability	25%	**4 (1.00)**	5 (1.25)	2 (0.50)	3 (0.75)
Feasibility	15%	**4 (0.60)**	4 (0.60)	1 (0.15)	3 (0.45)
TOTAL SCORE	100%	**4.00**	2.75	2.45	3.00

Park and Plant scores highest (4.00/5.0) — the most balanced and context-appropriate solution

RECOMMENDATION: EXPAND PARK AND PLANT

https://www.qf.org.qa/park-and-plant. (Qatar Foundation, 2026)

- **Already Operational**
- **Eco-Friendly**
- **Compounding Impact**
- **Climate-Proof**

Implementation Phases

1. **Phase 1 (Years 1–2)-** High traffic areas
2. **Phase 2 (Years 2–4)-** Low traffic areas
3. **Phase 3 (Years 4–6)-** 70% EC-wide coverage

QNV 2030: Environmental sustainability, human development, and 60–70% shade target

Trade Offs

Time Lag
3–5 years before meaningful canopy. Existing canopies stay up during growth.

Water Dependency
Scale depends on Ashghal partnership for treated wastewater irrigation.

Establishment Mortality
Newly planted trees vulnerable to extreme heat events in first summers.

THANK YOU

Questions?

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REFERENCES